Practice workbook · Finish an if/elif/else so each input gives the right message

A given program has an unfinished if statement; the student writes the conditions (relational operators, and/or/not, range boundaries) and branches so each input produces the output in a table or description.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-CODE-02

1. Question 3 · part c · 6 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part c only.

3 Programs use logic, display data and validate input.

(a) The truth table shows the initial values for A and B.

Complete the truth table to show the results of each operation.

(4)

A	B	A OR B	NOT B	(A OR B) AND (NOT B)
0	0			
0	1			
1	0			
1	1			

(b) **Figure 2** shows the intended output from a program that displays every other name in an array of star names.

```
Alasia
Castor
Electra
Gudja
Izar
Kang
Maia
Ogma
```

Figure 2

Open **Q03b** in the code editor.

Amend the code to display every other name in the array of star names.

Do not add any further functionality.

Save your code as **Q03bFINISHED** with the correct file extension for the programming language.

(3)

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P 7 5 7 3 7 A 0 6 1 6

1. Question 3 · part c · 6 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part c only.

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(c) A program validates a number input by the user.

Figure 3 shows the output messages based on the inputted number.

Input	Output
<empty>	You must provide a number
Any negative number	The number must be greater than zero
0	The number must be greater than zero
1 to 20	Acceptable
60 or more	Acceptable
31 to 39	Centre
30	Perfect
Any other number	No message

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Figure 3

Open **Q03c** in the code editor.

Amend the code to ensure the messages are generated correctly.

Do not add any further functionality.

Save your code as **Q03cFINISHED** with the correct file extension for the programming language.

(6)

(Total for Question 3 = 13 marks)

1. Question 3 · part c · 6 marks · Working and answer

Working and answer

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2. Question 1 · part f · 4 marks

2019 JUN · Paper 02

Further practice · Attempt part f only.

1 Computer programs use several different programming constructs.

(a) Identify the **most appropriate** data type for the value 4.5

(1)

- ☒ **A** Boolean
- ☒ **B** Character
- ☒ **C** Real
- ☒ **D** String

(b) Give a definition of a logic error.

(1)

(c) Open **Q01c** in the code editor.

The program should print out a counter and the counter with 7 added to it.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

(d) Give **one** reason why computer programmers should use comments in their programs.

(1)

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2. Question 1 · part f · 4 marks

2019 JUN · Paper 02

Further practice · Attempt part f only.

(e) Describe the difference between a local variable and a global variable in a computer program.

(2)

(f) Trevor is the manager of a shoe shop.

Sales assistants can earn a bonus based on the numbers of pairs of shoes they sell and the total income for the shop each day.

Open **Q01f** in the code editor.

Amend the code to complete the 'if statement' used to produce the outputs described in the table.

Condition	Output
Shop income is more than £5000 or sales assistant has sold at least 10 pairs of shoes	Bonus is 10% of salary
Shop income is £2000 or more and sales assistant has sold at least 5 pairs of shoes	Bonus is 5% of salary

Do not add any further functionality.

Save your amended code as **Q01fFINISHED** with the correct file extension for the programming language.

(4)

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P 6 1 8 8 1 R A 0 3 1 6

3

Turn over ▶

2. Question 1 · part f · 4 marks

2019 JUN · Paper 02

Further practice · Attempt part f only.

(g) Car registration codes consist of seven characters.

The characters are two upper case letters followed by two numbers and then three upper case letters.

YH62KLV, EY09VPB and CK53PRJ are valid codes.

All codes need to be validated.

Complete the table to show **two** additional validation tests.

For each test give **one** example of erroneous data. The example that you give should fail **only** that test.

(4)

Validation test	Erroneous data
Is it 7 characters?	AB12CD

(Total for Question 1 = 16 marks)

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2. Question 1 · part f · 4 marks · Working and answer

Working and answer

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3. Question 4 · part a · 6 marks

2020 NOV · Paper 02

Further practice · Attempt part a only.

4 Gianluca is a schoolteacher. He is in charge of a year group of pupils.

(a) He has started to create a program to analyse test results.

Open **Q04a** in the code editor.

Amend the code to complete the 'if statement' used to produce the outputs described in this table.

Condition	Output text
English test score was less than 40; Maths test score was less than 50	Student failed both tests
English test score was less than 40 or Maths test score was less than 50	Student failed one test
English test score was at least 80; Maths test score was at least 85	Student passed both tests with distinction
Other conditions	Student passed both tests

Do not add any further functionality.

Save your code as **Q04aFINISHED** with the correct file extension for the programming language.

(6)

(b) Gianluca has started to create a program to analyse pupil attendance.

He wants to use a menu system with subprograms.

Open **Q04b** in the code editor.

The program must include subprograms to:

- display the **names** of students whose percentage attendance was less than 75%
- count and display the **number** of students whose percentage attendance was 90% or higher.

Complete the program code to implement these requirements using the structure given in **Q04b**.

Do not add any further functionality.

Save your code as **Q04bFINISHED** with the correct file extension for the programming language.

(7)

(Total for Question 4 = 13 marks)

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3. Question 4 · part a · 6 marks · Working and answer

Working and answer

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4. Question 2 · part b · 4 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

2 A book club uses computer applications.

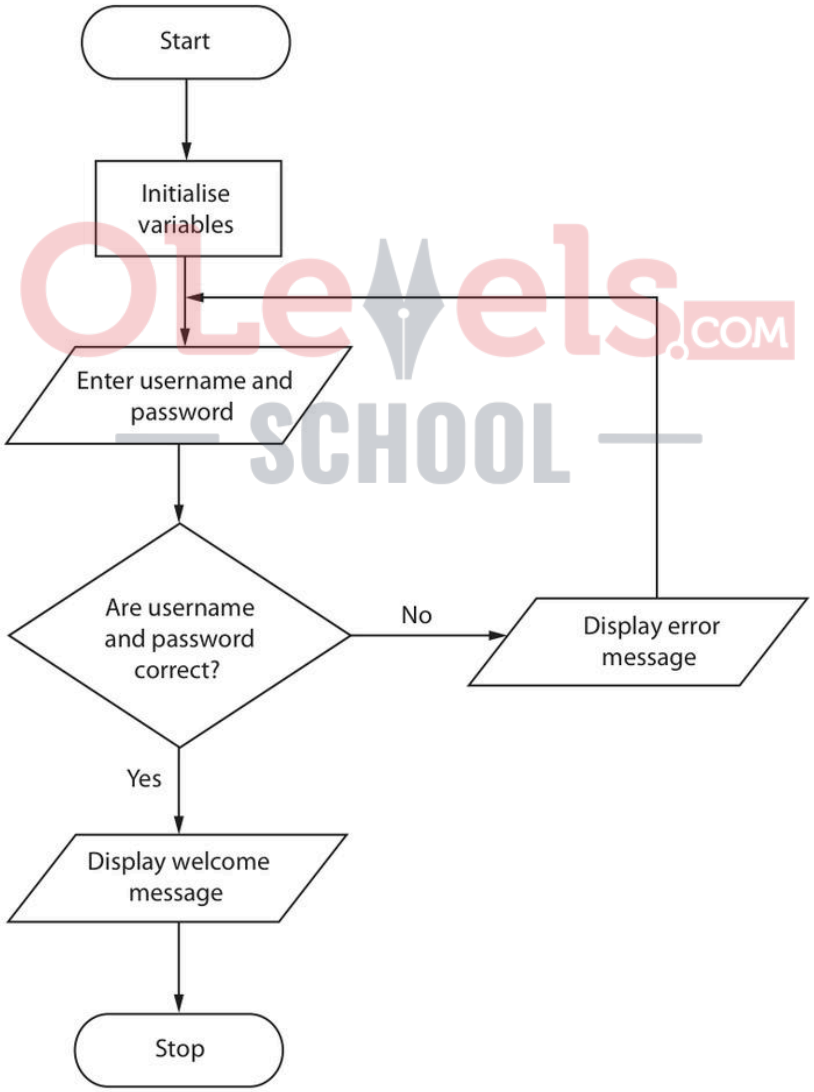
(a) The club wants a program to ensure that logins are valid.

A flowchart for a test version of the program has been designed.

The test version uses:

- a username of bard423
- a password of nX2934?

Here is the flowchart.



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4. Question 2 · part b · 4 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

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Write a program to implement the logic in the flowchart.

Open **Q02a** in the code editor.

You must use the structure given in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(11)

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4. Question 2 · part b · 4 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

(b) The club sells books.

It needs a computer program to monitor:

- the number of books sold
- the amount of profit made.

Q02b provides a structure for the program.

Open **Q02b** in the code editor.

Amend the code to complete the If statement used to produce the outputs described in the table.

Condition	Output message
Number of books sold is at least 5 and profit made is at least 10	Sales and profit are good this week
Number of books sold is over 20 and profit made is at least 20	Sales and profit are excellent this week
Number of books sold is under 5 or profit made is under 5	Poor performance this week
All other inputs	Alert manager

Do not add any further functionality.

Save your code as **Q02bFINISHED** with the correct file extension for the programming language.

(4)

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P 6 6 4 9 0 A 0 6 1 6

4. Question 2 · part b · 4 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

(c) Boundary testing is an important part of program development.

The boundary value is the most extreme value that will pass the test.

For example, if the boundary condition is no more than 30 then the boundary test data would be 30.

Complete the test table to show the boundary test data that could be used for each test.

(3)

	Test data	Expected output
booksSold		Poor performances this week
profit		
booksSold		Sales and profit are good this week
profit		
booksSold		Sales and profit are excellent this week
profit		

(Total for Question 2 = 18 marks)

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4. Question 2 · part b · 4 marks · Working and answer

Working and answer

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5. Question 1 · part g · 4 marks

2022 JUN · Paper 02

Further practice · Attempt part g only.

1 Programmers write code to solve problems.

(a) Identify the description of a variable in a computer program.

(1)

- ☒ **A** A value that cannot be used more than once
- ☒ **B** A value that must be input
- ☒ **C** A value that is always used in a calculation
- ☒ **D** A value that can change

(b) Identify the technique that improves the readability of code.

(1)

- ☒ **A** Using indents on every line
- ☒ **B** Using descriptive names for variables
- ☒ **C** Using the correct operators
- ☒ **D** Using suitable data structures

(c) Complete the table by giving an example value for **each** data type.

The first row has been completed for you.

(2)

Data type	Example value
integer	12
char	
real	

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5. Question 1 · part g · 4 marks

2022 JUN · Paper 02

Further practice · Attempt part g only.

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(d) Describe **one** difference between the data used in boundary testing and the data used in erroneous testing.

(2)

(e) A program should output the value 2

However, there is an error in the code and the actual output is 4

Name this type of error.

(1)

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(f) Open **Q01f** in the code editor.

The program should allow the input of the length of the side of a square and output the area of the square.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01fFINISHED** with the correct file extension for the programming language.

(3)

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(g) A program is needed that will accept the input of a letter and compare it with a stored letter.

It will check whether the letter comes earlier in the alphabet, later in the alphabet or is the same letter as the stored letter.

It will output the letter and the stored letter with an appropriate message.

Open **Q01g** in the code editor.

Amend the code to complete the if statement used to produce the output.

You must use the structure given in **Q01g** to complete the program.

Do not add any further functionality.

Save your code as **Q01gFINISHED** with the correct file extension for the programming language.

(4)

(Total for Question 1 = 14 marks)

5. Question 1 · part g · 4 marks · Working and answer

Working and answer

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6. Question 1 · part b · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.

1 Programmers write programs to solve problems.

(a) Programmers use many different techniques and tools to solve problems.

Identify what is meant by the term **abstraction**.

(1)

- ☒ **A** Breaking a problem down into smaller more manageable parts
- ☒ **B** Deciding if a program is efficient in terms of execution time
- ☒ **C** Drawing flowcharts with symbols used in the computer industry
- ☒ **D** Removing unnecessary detail to highlight the important points

(b) Open **Q01b** in the code editor.

A user enters a name and an age when the program executes.

The program should display a welcome message when the user is less than 30 years of age.

Amend the code to complete the program.

Save your amended code as **Q01bFINISHED** with the correct file extension for the programming language.

(4)

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6. Question 1 · part b · 4 marks

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Further practice · Attempt part b only.

(c) Open **Q01c** in the code editor.

Use the code to answer these questions.

(i) Give the text of a comment used in the code.

(1)

(ii) Give the text of a line that creates and initialises a variable.

(1)

(iii) Give the keyword that starts the selection used in the code.

(1)

(iv) Give the logical operator used in the code.

(1)

(d) Arrays and records are data structures.

Identify the data that **must be** stored in a record rather than an array.

(1)

- ☒ **A** "King Lear", "Macbeth", "Romeo and Juliet", "The Tempest"
- ☒ **B** "King Lear", 1983, 78.32, 99
- ☒ **C** 1978, 1985, 1990, 2001
- ☒ **D** 13.99, 12.75, 10.58, 11.43

(e) Complete the table to show the data type for **each** item.

(2)

Item	Data type
45.82	
True	

(Total for Question 1 = 12 marks)

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6. Question 1 · part b · 4 marks · Working and answer

Working and answer

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